# Diego Macías Vega

Final-year science student in Madrid, applying for 2027 entry to Aerospace Engineering, Applied Physics and Advanced Technology

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## PROFILE

I build software on physics and mathematics and check it against measurement: reproduced baselines, validation suites and published negative results. I work with AI coding assistants and say so: the code is largely AI-written; the questions, the rules each project obeys, the decisions and the responsibility for every claim are mine.

## EDUCATION

**Bachillerato in Sciences and Technology** IES San Agustín del Guadalix, Madrid — 2025 – 2027

- Mathematics II, Physics, Technology and Engineering, Computer Science. Final year; graduating June 2027.

**University applications, September 2027 entry** the Netherlands — in preparation

- BSc Aerospace Engineering, TU Delft (selection exam) · BSc Applied Physics, TU Eindhoven · BSc Advanced Technology, University of Twente.

**CS50x: Introduction to Computer Science** Harvard University, online — 2026, in progress

## PROJECTS

**RAAN · Open Orbital Intelligence Lab** Python, TypeScript, WebGL, LightGBM — 2026

- Reproducible lab on public orbital data. Digital twin of 20,152 catalogued objects propagated with SGP4; propagator checked against Vallado's 666 reference vectors (max difference 3.5e-5 km).
- ESA Collision Avoidance Challenge: naive forecast reproduced exactly (L = 0.694). Best model scores 0.627 on the test set but fails cross-validation; reported as a negative result.
- Manoeuvre detection on 15 satellites, leave-one-satellite-out: F1 0.81 (classifier), 0.74 (physics baseline), 0.09 (chance). 6-page technical report.
- Code mostly written by an AI coding assistant and reviewed by an independent session; I set the question and the rules and answer for every claim.

**QEC Lab** quantum error correction — 2026, in progress

- Open lab comparing neural decoders with minimum-weight perfect matching on the surface code. Under construction; no results yet.

**gravity-lab** C++17, numerical methods — 2026

- N-body simulator of the Solar System validated against measurement: reproduces Mercury's relativistic perihelion precession (42.81 arcsec/century; accepted 42.98, error 0.4 %).
- Four integrators compared; Yoshida 4 (symplectic) holds energy error at 1.4e-11 over 100 years. 31 validation cases in CI. Browser build online.

**KAIROS** Python, FastAPI, PostgreSQL + pgvector, Docker — 2026

- Local-first personal AI assistant: 17 agents, 300+ tests, sandboxed self-modification that never merges without my approval. Architecture and decisions mine (77 written decision records); Claude as pair programmer.

**Other** — 2026

- Chess trainer (Stockfish and Maia, explanations fact-checked by test) · Quant Path (drills and 110 code-verified problems for quantitative trading) · TraceGuard (firmware changes checked against safety rules; paused MVP) · Sympto+ (multilingual health-information web app; paused).

## EXPERIENCE

**Private tutor** Madrid — ongoing

- One-to-one tuition for school students.

## SKILLS, LANGUAGES AND OTHER

**Technical:** Python, C++17, TypeScript; numerical integration, orbital propagation (SGP4), gradient boosting and recurrent networks, evaluation design (baselines, cross-validation, bootstrap); Git, Linux/WSL, Docker, GitHub Actions.

**Languages:** Spanish (native) · English, Cambridge C1 Advanced (score 186, obtained at 16) · Chinese, YCT 2.

**Sport:** Judo black belt; Madrid champion at age 12 (no longer competing).